

Sampling distributions,
standard errors, and confidence intervals

Overview

Very Quick review: Sampling distributions and the standard error

Exploring sampling distributions in R

Confidence intervals

Announcement

Homework 4 is due on Gradescope on **Sunday October 4th at 11pm**

- **Be sure to mark each question on Gradescope!**

The material in the coming week is going to be a bit more conceptually challenging

Please attend the practice sessions and office hours to reinforce your understanding!

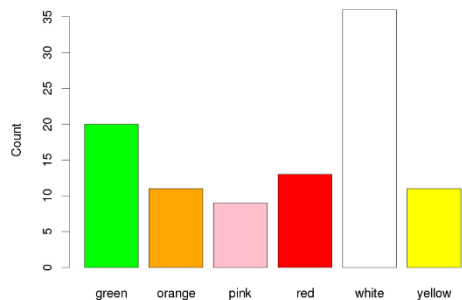
Review: Sampling distributions

Sampling distribution

THEORETICAL CONCEPT

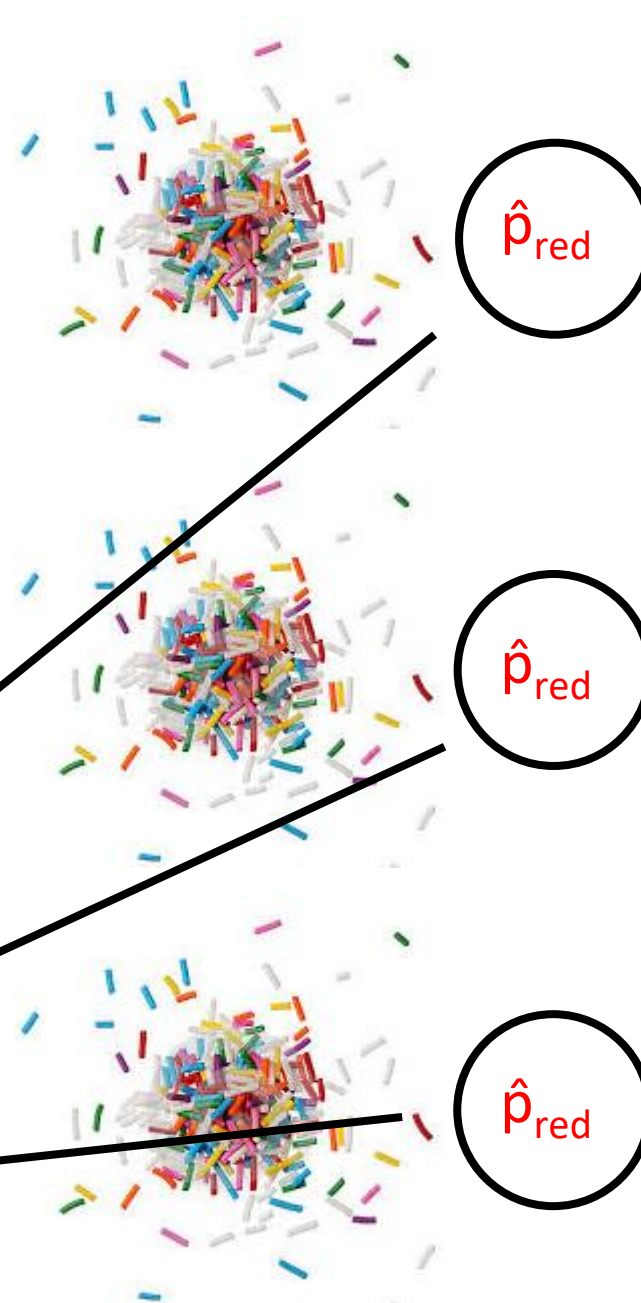
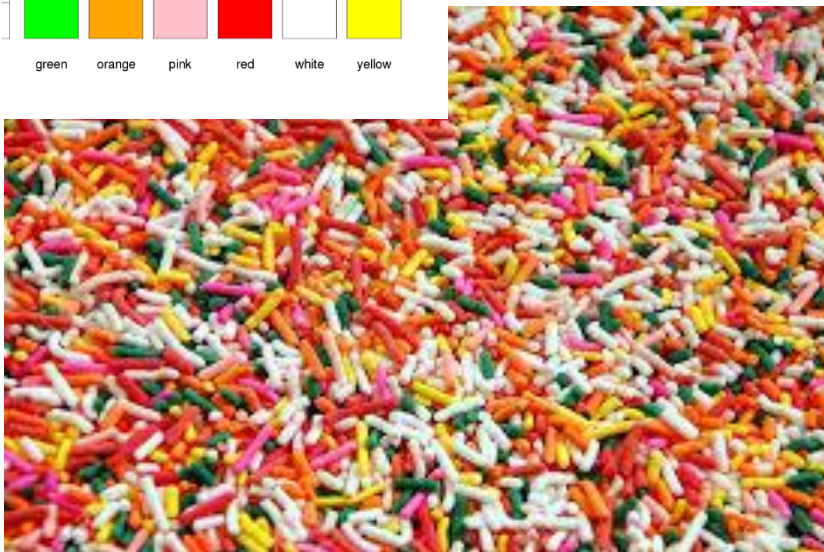
A **sampling distribution** is the distribution of sample statistics computed from different samples of the same size (n) from the same population

A sampling distribution shows us how the sample statistic varies from sample to sample

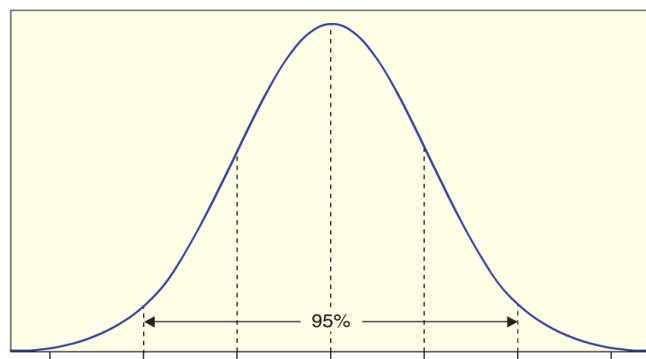


π_{red}

$n = 100$

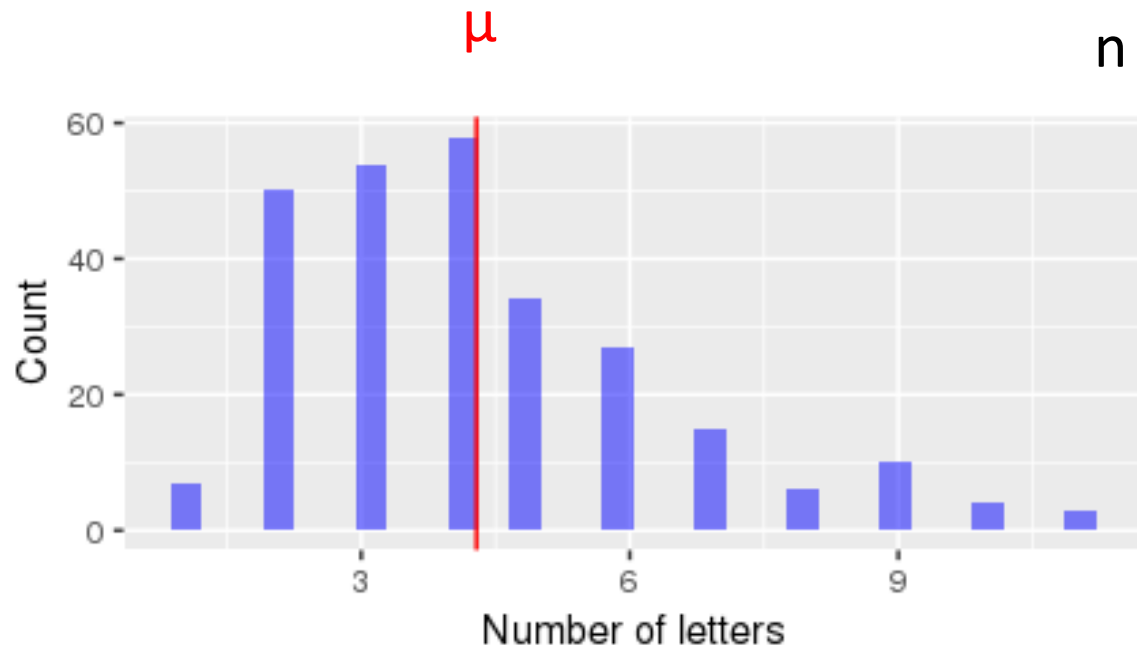


Distribution
of $\hat{p}$'s



Sampling distribution!

Gettysburg address word length sampling distribution



10, 3, 3, 3, 4,
3, 2, 6, 10, 5

$$\bar{x} = 5$$

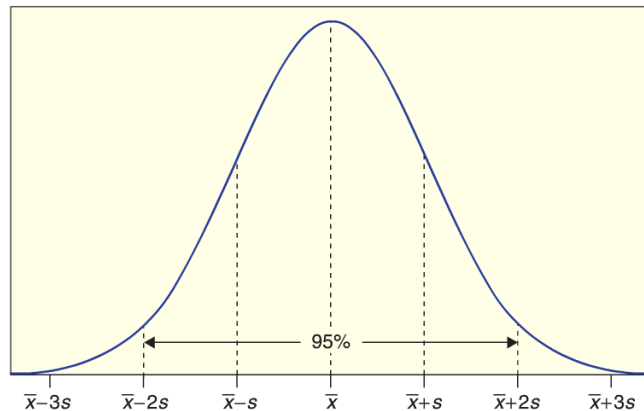
2, 6, 2, 6, 6,
2, 5, 3, 2, 9

$$\bar{x} = 4.3$$

3, 9, 3, 4, 4,
3, 6, 6, 2, 2

$$\bar{x} = 4.2$$

Distribution
of $\bar{x}$'s



Sampling distribution!

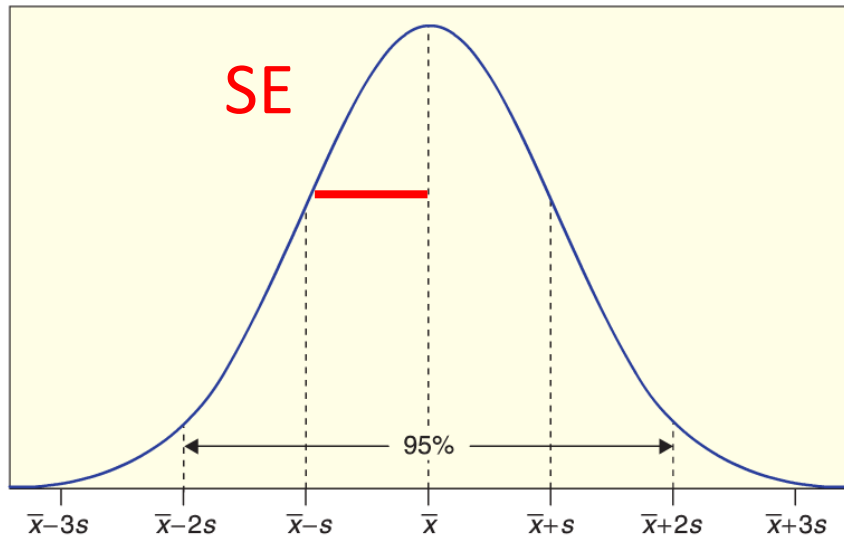
[Gettysburg sampling distribution app](#)

The standard error

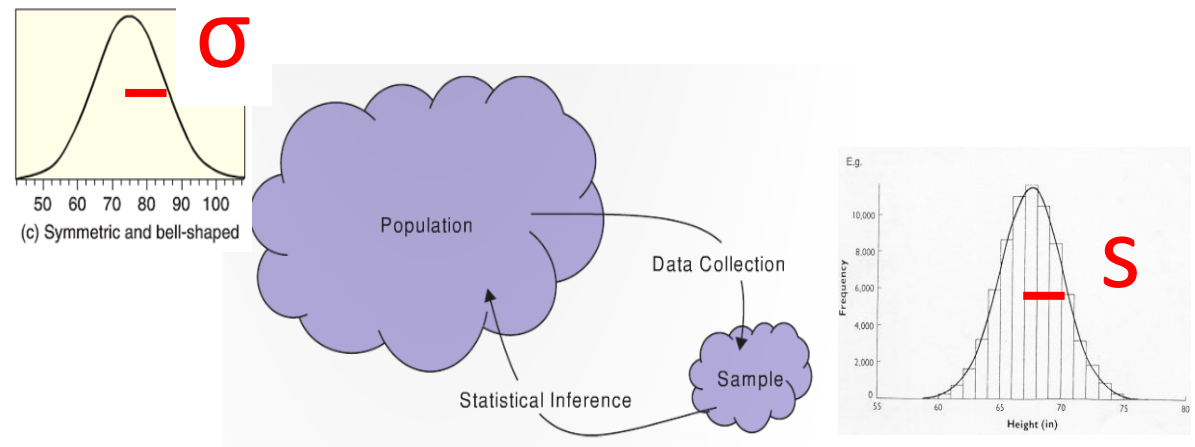
The **standard error** of a statistic, denoted SE, is the standard deviation of the sample statistic

- i.e., SE is the standard deviation of the *sampling distribution*

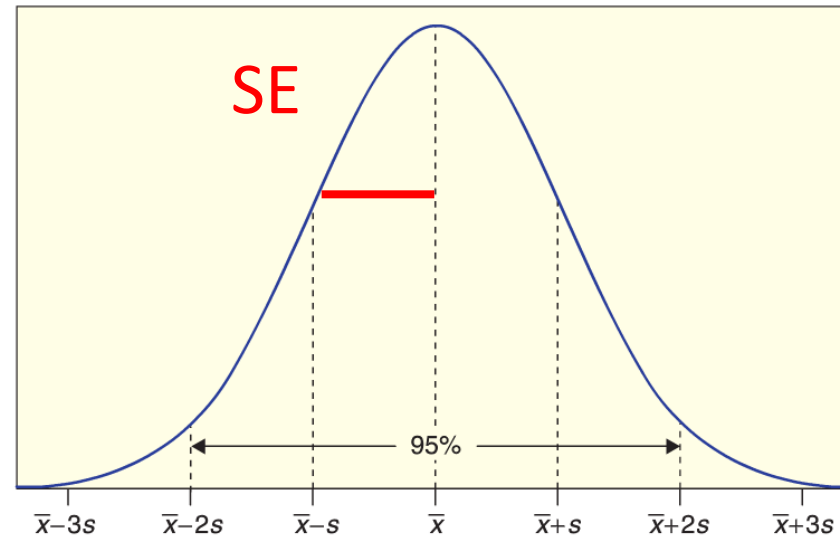
Note: standard error SE is different from σ and s !



Sampling distribution!



What does the size of a standard error tell us?



Q: If we have a large SE, would we believe a given statistic is a good estimate for the parameter?

- E.g., would we believe a particular $\bar{x}$ is a good estimate for μ ?

A: A large SE means our statistic (point estimate) could be far from the parameter

- E.g., $\bar{x}$ could be far from μ

Sampling distributions in R!

Let's create a sampling distribution in R

Load the SDS1000 library to make all SDS1000 functions available

```
library(SDS1000)
```

Get the Gettysburg population data

```
load("gettysburg.Rda")
```

```
word_lengths <- gettysburg$num_letters    # lengths of the 268 words
```

Let's create a sampling distribution in R

We can use the `sample(data_vec, n)` to get a sample of length n:

```
curr_sample <- sample(word_lengths, 10)
```

Q: How can we get $\bar{x}$ from this sample in R?

```
mean(curr_sample)
```

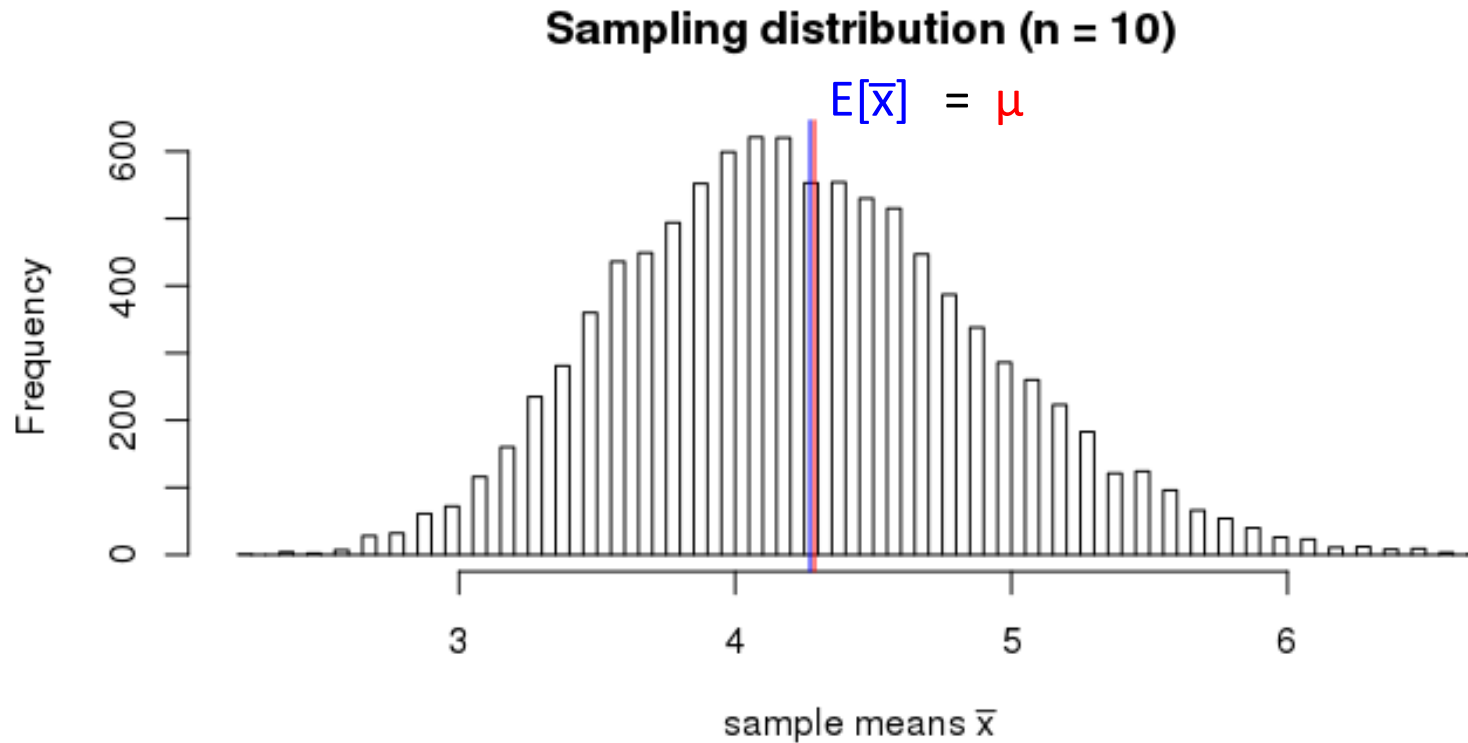
Q: How could we get a full sampling distribution?

- A: Repeat this many times to get an approximation of the sampling distribution
- If we store the $\bar{x}$'s in a vector, we can then plot the sampling distribution as a histogram

Let's create a sampling distribution in R

```
sampling_dist <- do_it(10000) * {  
  
  curr_sample <- sample(word_lengths, 10)  
  mean(curr_sample)  
  
}  
  
hist(sampling_dist)  
SE <- sd(sampling_dist)
```

Sampling distribution in R



```
mean(sampling_dist)
```

```
mean(word_lengths) # these are the same, so no bias
```

Changing the sample size n

What happens to the sampling distribution as we change n ?

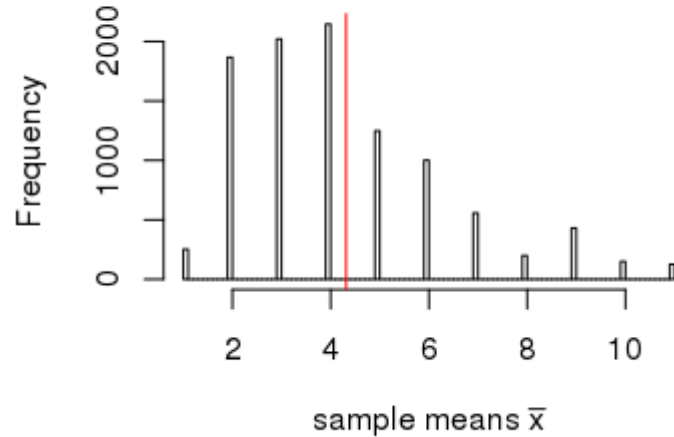
- Experiment for $n = 1, 5, 20, 80$

```
sampling_dist <- do_it(10000) * {  
    curr_sample <- sample(word_lengths, 20)  
    mean(curr_sample)  
}
```

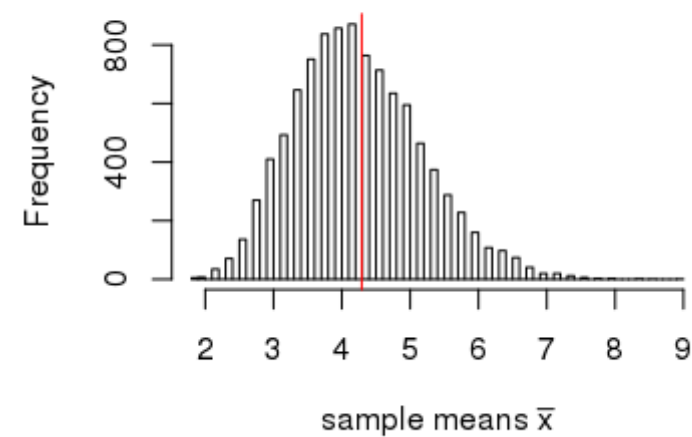
```
hist(sample_means, breaks = 100)
```

[Gettysburg sampling distribution app](#)

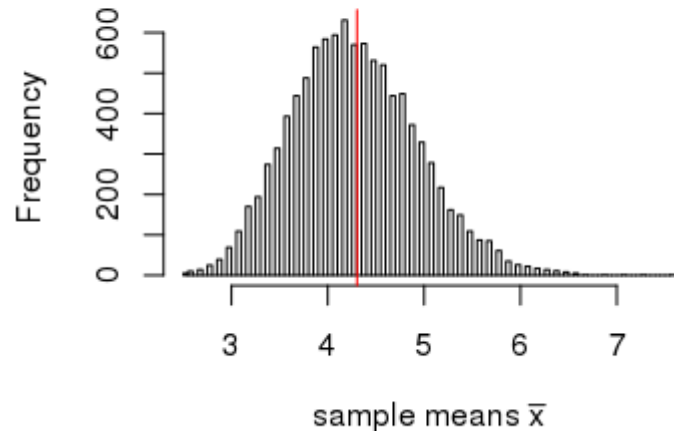
Sampling distribution ($n = 1$)



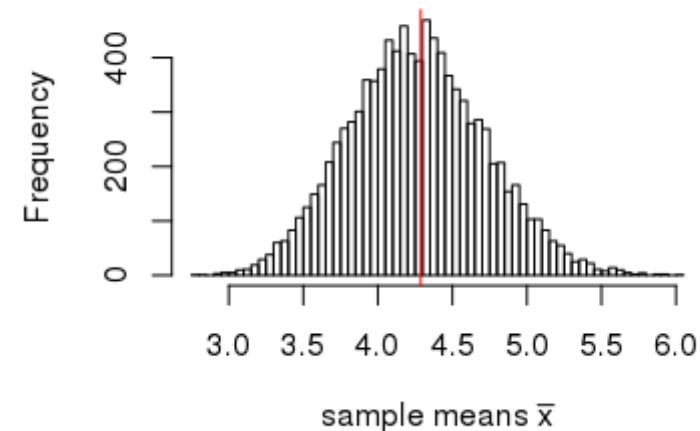
Sampling distribution ($n = 5$)



Sampling distribution ($n = 10$)



Sampling distribution ($n = 20$)



x-axis range 9 vs. 6

As the sample size n increases

1. The sampling distribution becomes more like a normal distribution
2. The sampling distribution points ($\bar{x}$'s) become more concentrated around the mean $E[\bar{x}] = \mu$

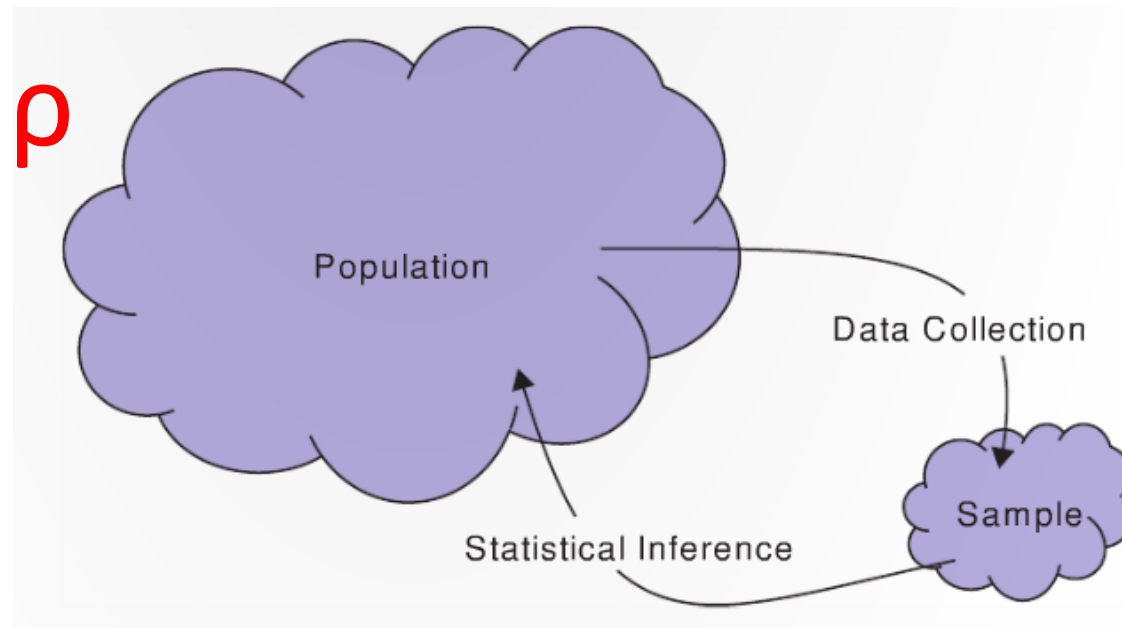
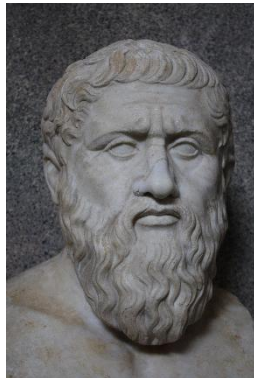
Point estimates and confidence intervals

Back to the big picture: Inference

Statistical inference is...?

the process of drawing conclusions about the entire population based on information in a sample

π, μ, σ, ρ



$\hat{p}, \bar{x}, s, r$



Point Estimate

We use a statistic from a sample as a **point estimate** for a population parameter

- $\bar{x}$ is a point estimate for...? μ

40% of Americans approve of Trump's job performance according to a poll of 2,464 people

Q: What are π and $\hat{p}$ here?

Q: Is $\hat{p}$ a good estimate for π in this case?

A: We can't tell from the information given



([outdated graphic](#))

Interval estimate based on a margin of error

An **interval estimate** give a range of plausible values for a population parameter

One common form of an interval estimate is:

$$\textit{Point estimate} \pm \textit{margin of error}$$

Where the **margin of error** is a number that reflects the precision of the sample statistic as a point estimate for this parameter

Example: Gallup poll

40% of American approve of Trump's job performance, plus or minus 3%

How do we interpret this?

Says that the population parameter (π) lies somewhere between 37% to 43%

i.e., if they sampled all voters, the true population proportion (π) would likely be in this range

Confidence Intervals

A **confidence interval** is an interval computed by a method that will contain the *parameter* a specified percent of times

- i.e., if the estimation were repeated many times, the interval will have the parameter x% of the time

The **confidence level** is the percent of all intervals that contain the parameter

Think ring toss...

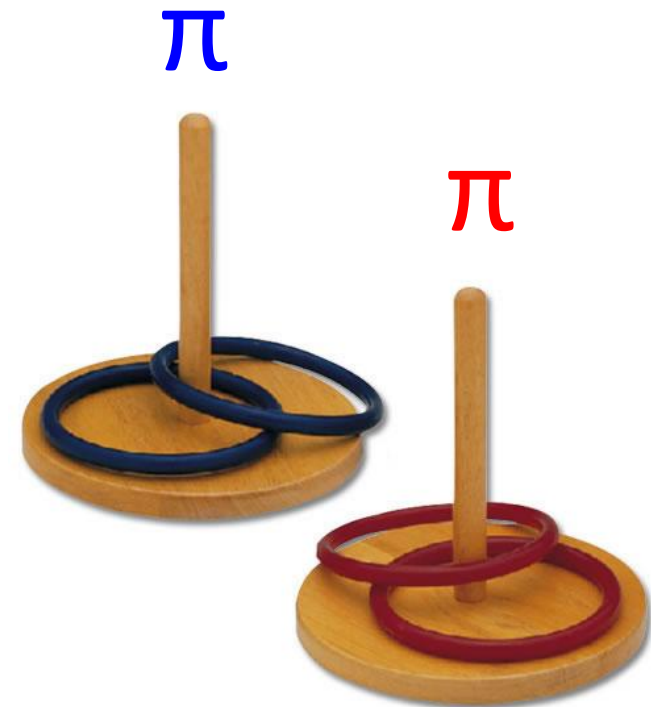
A parameter exists in the world

We toss intervals at it

- These are our confidence intervals

95% of those intervals capture the parameter

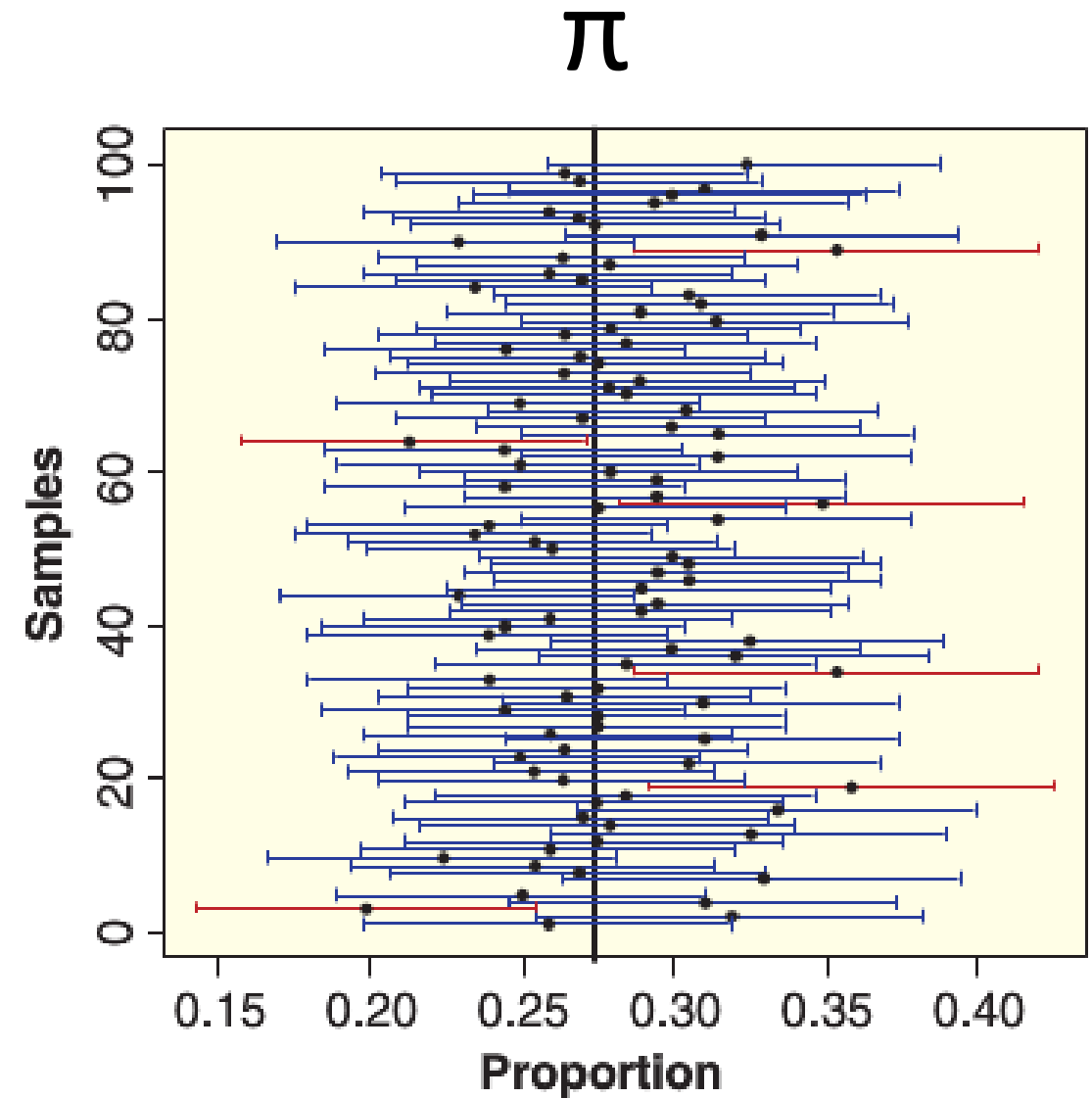
- (95% is our confidence level)



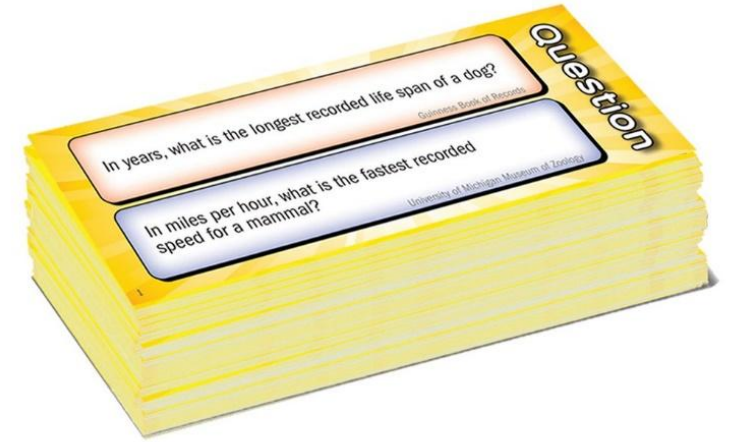
Confidence Intervals

For a **confidence level** of 95%...

95% of the **confidence intervals** will have the parameter in them



Wits and Wagers: 90% confidence interval estimator



I will ask 10 questions that have numeric answers

Please come up with a range of values that contains the true value for 9 out of the 10 questions

- I.e., be a 90% confidence interval estimator

Write your range of estimates for each question as two numbers

- E.g., [10.2 to 50.7]

Confidence intervals

Congratulations, if you are all 90th confidence interval estimators you got 9 out of 10 questions correct!

Unfortunately, you don't know which one you got wrong

But if you are 90% confidence interval estimators, you know you were right most of the time!



100% confidence intervals

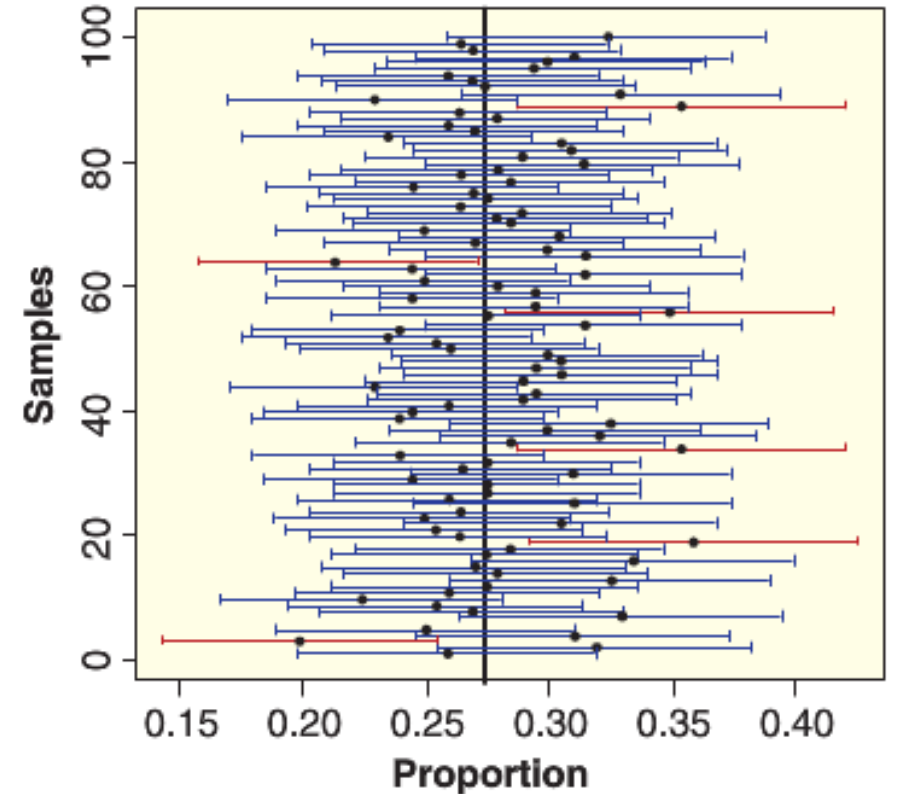
There is a tradeoff between the **confidence level** (percent of times we capture the parameter) and the **confidence interval size**

Note

For any given confidence interval we compute, we don't know whether it has really captured the parameter

But we do know that if we do this 100 times, 95 of these intervals will have the parameter in it

(for a 95% confidence interval)



Preview: Computing confidence intervals

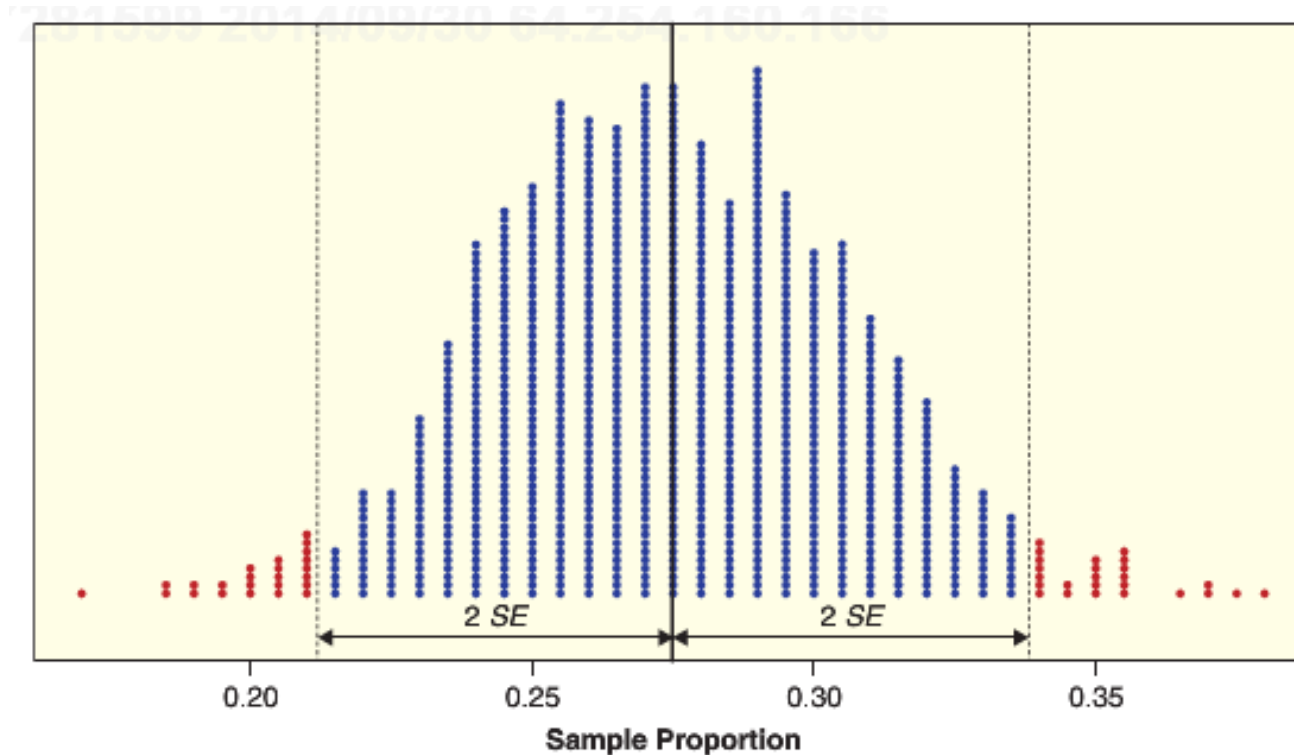
We will learn how to compute confidence intervals over the next few classes, but first let's review some material...

Spoiler: they will have the form:

$$\text{Confidence interval} = \text{statistic} \pm 2 \cdot \text{Standard Errors}$$

Let's preview why this is the case...

Recall: Sampling distributions are usually normal



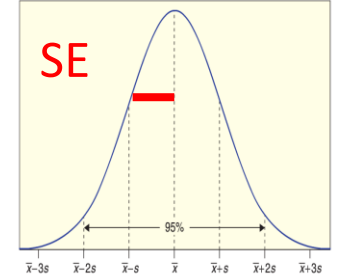
So 95% of the sample statistics will fall within ± 2 Standard Errors from the mean

Thus, if we knew the SE, we could calculate a 95% confidence interval!

- Why is this true?

The problem

Unfortunately, we don't know the Standard Error 😞



Sampling distribution!

Q: Could we repeat the sampling process many times to estimate it?

A: No, we can't do an experiment that many times

We're just going to have to pick ourselves up from the bootstraps!

Estimate SE with $\hat{SE}$

Then use $\bar{x} \pm 2 \cdot \hat{SE}$ to get the 95% CI

Next class

Using the bootstrap to estimate the SE...

